

Automated Image Captioning for Web Accessibility Using an Encoder-Decoder Transformer Architecture

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+12.8×BLEU-4 vs Baseline
0.1625 vs 0.0127**+26.9%**CLIPScore vs Baseline
0.7753 vs 0.6111**0.3296**ROUGE-L
↑ +6.4% over baseline**0.5276**BLEU-1
on par with baseline

Problem Statement

- A vast portion of the internet's visual content lacks descriptive alt-text, creating an accessibility barrier for visually impaired users who rely on screen readers
- Manual captioning is time-consuming and frequently overlooked by content creators
- Automating this process can efficiently make digital platforms more inclusive
- **Goal:** Develop a neural network that processes an image and generates a coherent, contextually accurate natural language description of its contents

Dataset: Flickr8k

8,000
Images5×
Captions each40K
Total pairs

- Split: 6,000 train / 1,000 val / 1,000 test
- Input: RGB images resized & normalized (224×224)
- Output: sequence of text tokens → descriptive sentence

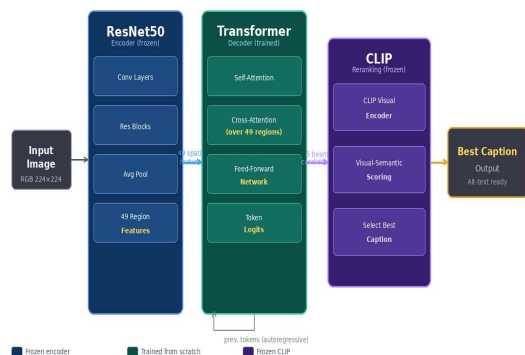
Training Methodology

- Train Transformer encoder-decoder from scratch on Flickr8k (6K/1K/1K split)
- Frozen ResNet50 backbone extracts 49 spatial region features (7×7 grid)
- Multi-head Transformer decoder attends over spatial regions to generate captions
- Inference: beam search (5 candidates) + CLIP-guided reranking for best caption
- Baseline: CNN+LSTM with single pooled vector and greedy decoding

Evaluation Metrics

BLEU-1/4	Precision of n-gram overlap vs. reference captions
ROUGE-L	Longest common subsequence — measures fluency & flow
CIDEr	TF-IDF weighted; rewards unique image-specific descriptions
CLIPScore	Vision-language alignment via CLIP embeddings (our addition)

ENCODER-DECODER Pipeline



Architecture Overview



Main References

- [1] Flickr8k Dataset. Kaggle: <https://www.kaggle.com/datasets/adityajn105/flickr8k>
- [2] Vaswani, A. et al. (2017). Attention is All You Need. NeurIPS 2017.
- [3] Radford, A. et al. (2021). Learning Transferable Visual Models from Natural Language Supervision (CLIP). ICML 2021.

Cross-attention detail – CLIP reranking detail



Model vs. Baseline — All Metrics

